

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	D	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,D	A,B,C,D	A,D	A,B,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.10	110.00	15.00	7.50	0.25	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	63	5	8			

PART-2 : CHEMISTRY

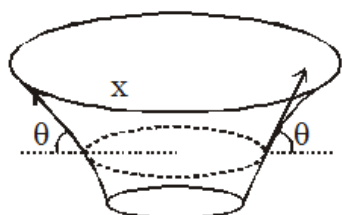
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	A	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,C,D	A,C	A,B,D	A,C,D	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	400.00	100.00	2.66 to 2.67	2.00	3.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	2	6	7			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	B	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,C	A	B,C,D	A,C	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	3.00	6.00	6.00	10.00	2.82 or 2.83
SECTION-II (ii)	Q.	7	8	9			
	A.	4	5	7			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

2. Ans (A)



$$2\pi x \sigma \times 2 \sin \theta = mg$$

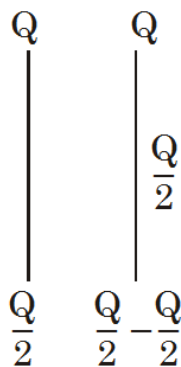
$$\sin \theta = \frac{mg}{4\pi x \sigma} \ll 1 \Rightarrow \sin \theta = \tan \theta$$

$$\frac{dy}{dx} = \frac{mg}{4\pi x \sigma}$$

$$dy = \frac{mg}{4\pi \sigma} \int_R^{\frac{D}{2}} \frac{dx}{x}$$

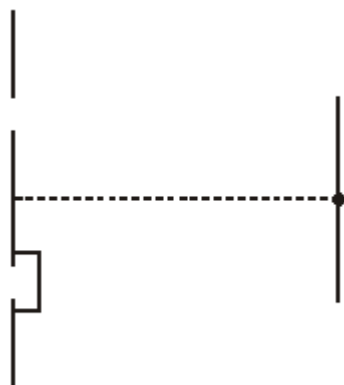
$$y = \frac{mg}{4\pi \sigma} \ln \left(\frac{D}{2R} \right)$$

3. Ans (D)



$$U = \frac{\left(\frac{Q}{2}\right)^2}{2C} = \text{heat dissipated}$$

4. Ans (B)



$$4I_0 = 4 \text{ W/m}^2$$

$$I_0 = 1 \text{ W/m}^2$$

$$\Delta\phi = \frac{2\pi}{\lambda}(\mu - 1)t$$

$$= \frac{2\pi}{600} \times 0.5 \times 200 = \frac{\pi}{3}$$

$$I = I_0 + \eta I_0 + 2\sqrt{\eta} I_0 \cos \frac{\pi}{3}$$

$$1.75 = 1 + \sqrt{\eta} + \eta$$

$$\sqrt{\eta} = x$$

$$x^2 + x - \frac{3}{4} = 0$$

$$x = \frac{-1 + \sqrt{1+3}}{2} = \frac{1}{2}$$

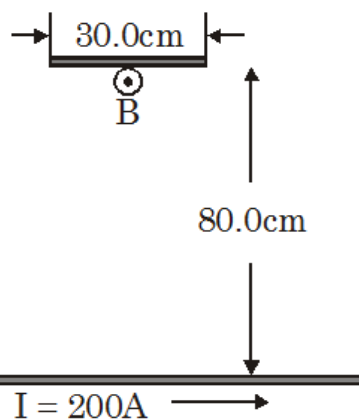
$$\Rightarrow \eta = \frac{1}{4}$$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = \left(1 + \frac{1}{\sqrt{4}}\right)^2 = 2.25 \text{ W/m}^2$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C)



$$\varepsilon = B_{\perp} \ell v$$

$$v = gt$$

$$= 2 \text{ m/s}$$

$$S = \frac{1}{2}gt^2$$

$$= 5 \times 0.04$$

$$= 0.2$$

$$B_{\perp} = \frac{\mu_0 \times 200}{2\pi \times 0.6} \Rightarrow \varepsilon = \frac{4}{6} \times 10^{-4} \times 0.3 \times 2$$

$$= 0.04 \text{ mV}$$

6. Ans (A,D)

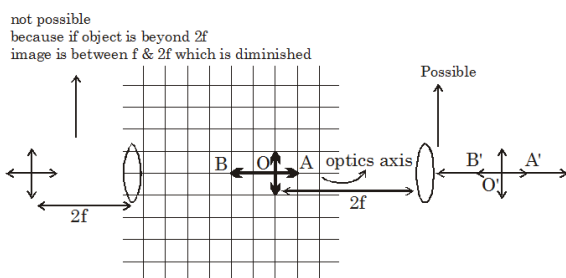
$$i = \frac{E}{r + R + r_1}$$

$$\frac{di}{dR} = \frac{-E}{(r + R + r_1)^2} = -\frac{i^2}{E}$$

$$\Rightarrow dR = \frac{diE}{i^2}$$

$$i_{\max} \Rightarrow dR_{\min}$$

7. Ans (A,B,C,D)



m at $0 = -1 \Rightarrow 0$ is at $2f$

$$x_0 x_I = -f^2$$

$$+ (f - a)(f + 2a) = +f^2$$

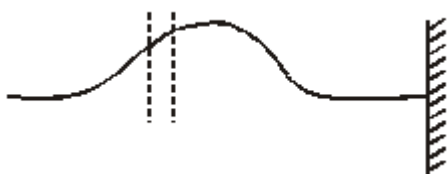
$$f^2 + 2af - af - 2a^2 = f^2$$

$$af = 2a^2$$

$$f = 2a$$

Real object & real image $\Rightarrow$ converging

8. Ans (A,D)



$$dU = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2 dx$$

$$dk = \frac{1}{2} \mu v_p^2 dx$$

At extreme $\frac{\partial y}{\partial x}$ & $v_p = 0$

at mean $\frac{\partial y}{\partial x}$ & v_p are both maximum

9. Ans (A,B,D)

$C = \text{constant} \Rightarrow$ polytropic

$$15 = C_v - \frac{R}{m-1}$$

$$15 = \frac{3}{2} \times \frac{25}{3} - \frac{25}{3(m-1)}$$

$$-2.5 = \frac{25}{3(m-1)}$$

$$m-1 = \frac{-10}{3} \Rightarrow m = -\frac{7}{3}$$

$$20 = W = \frac{nR(T_1 - T_2)}{m-1} = \frac{-1 \times \frac{25}{3} \Delta T}{-\frac{10}{3}}$$

$$\Delta T = 8k$$

$$\Delta H = W + nC_v \Delta T = 20 + 1 \times \frac{3}{2} \times \frac{25}{3} \times 8$$

$$= 120 \text{ J}$$

$PV^{-7/3} = C$ gas does work

$$\Rightarrow V \uparrow \Rightarrow P \uparrow$$

10. Ans (A,B,D)

$$-\mu mg \ell = \Delta k$$

$$-\mu \times 20 \ell = -500 \text{ J}$$

$$\mu \ell = 25$$

$$\frac{1}{2} \times 2 \times u^2 = 900 \Rightarrow u = 30 \text{ m/s}$$

$$\frac{1}{2} \times 2 \times v^2 = 400 \Rightarrow v = 20 \text{ m/s}$$

$$20 = 30 - \mu g \times 2$$

$$\Rightarrow \mu = 0.5$$

$$\ell = 50 \text{ m}$$

$$-\mu mg \times S = -\frac{1}{2} mv^2$$

$$-0.5 \times 10 \times S = \frac{1}{2} \times 400$$

$$S = 40 \text{ m}$$

$$0 = 20 - 5t \Rightarrow t = 4 \text{ sec.}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.10)

$$\frac{2}{5} MR^2 \times \frac{2\pi}{24} = \left(\frac{2}{5} MR^2 + mR^2 \right) \times \frac{2\pi}{30}$$

$$\Rightarrow \frac{5}{4} = \frac{5}{2} \frac{m}{M} + 1$$

$$\Rightarrow \frac{1}{4} = \frac{5}{2} \frac{m}{M} \Rightarrow \frac{m}{M} = 0.1$$

2. Ans (110.00)

$$0 = M \times v_c = -0.1 M v_a$$

$$v_a = 10 v_c$$

$$+ \frac{GMm}{R} = \frac{1}{2} m v_c^2 + \frac{1}{2} \times 0.1 M v_a^2$$

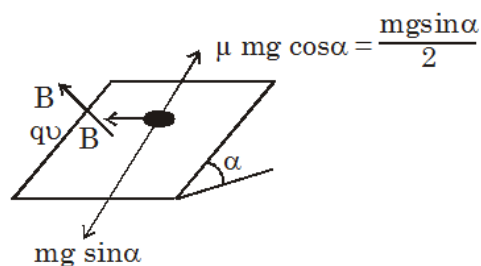
$$+ \frac{GM \times 0.1M}{R} = \frac{1}{2} M \times \frac{v_a^2}{100} + \frac{1}{2} \frac{M v_a^2}{10}$$

$$\frac{2GM}{R} = \frac{11 v_a^2}{10}$$

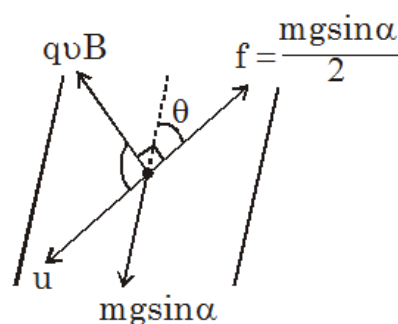
$$v_a = \sqrt{\frac{2GM}{R} \times \frac{10}{11}} = 11 \times \sqrt{\frac{10}{11}} = \sqrt{110} \text{ km/s}$$

3. Ans (15.00)

$$v = \frac{mg}{qB} \sqrt{\sin^2 \alpha - \mu^2 \cos^2 \alpha} = 15 \text{ m/s}$$



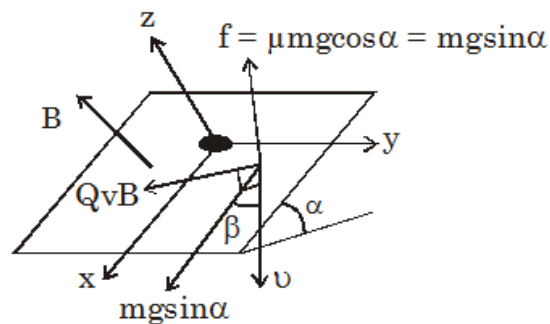
After a long time



$$(qvB)^2 + \left(\frac{mg \sin \alpha}{2} \right)^2 = (mg \sin \alpha)^2$$

$$v = \frac{mg \sin \alpha}{qB} \times \frac{\sqrt{3}}{2} = \frac{80 \times 10 \times 10 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}}{10^{-1} \times 0.4} = 15 \text{ m/s}$$

4. Ans (7.50)



Condition to come out of plane is that $y_{\max} \geq L$

Let us assume that at any instant velocity makes an $\angle \beta$ with line of greatest slope.

$$m a_x = mg \sin \alpha - q v_y B - mg \sin \alpha \cos \beta$$

along the direction of motion

$$\frac{m dv}{dt} = mg \sin \alpha \cos \beta - mg \sin \alpha$$

$$\Rightarrow m \frac{dv_x}{dt} + m \frac{dv}{dt} = -qB \frac{dy}{dt}$$

$$\Rightarrow v_x + v = -\frac{qBy}{m} + \text{constant}$$

Let's say disc is initially at origin, with velocity

u_0 at $\angle \theta$ to x-axis.

$$\Rightarrow v_0 + v_0 \cos \theta = \text{constant}$$

$$\Rightarrow v_x + v + \frac{qB y}{m} = v_0 (1 + \cos \theta)$$

at $y_{\max}$, $v_y = 0$ & $v_x = 0$

$$\Rightarrow y_{\max} = \frac{v_0}{qB} (1 + \cos \theta). \text{ For } y \text{ to be max } \theta = 0^\circ$$

$$\Rightarrow v_0 = \frac{qBL}{2m}$$

5. Ans (0.25)

$$12\text{gm} \rightarrow 6 \times 10^{23}$$

$$1\text{gm} \rightarrow \frac{6}{12} \times 10^{23} \times \frac{1}{7} \times 10^{11} = N_0$$

$$\frac{dN}{dt} = \frac{1}{2} \times \frac{10^{12}}{7} \times \frac{\ln 2}{2 \times 10^{11}} = \frac{1}{4} = 0.25$$

6. Ans (2.00)

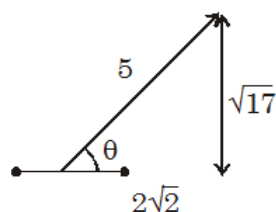
$$A = A_0 e^{-\lambda t}$$

$$t = 2T_{1/2} = 2 \times 10^{11} \text{ sec.}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (63)

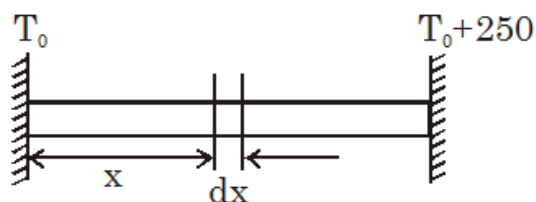


$$E = \frac{kp}{r^3} \sqrt{1 + 3\cos^2\theta}$$

$$= \frac{9 \times 10^9 \times 2.5 \times 10^{-6} \times 0.125 \times 10^{-3} \times 2}{5^3} \sqrt{1 + 3 \times \frac{8}{25}}$$

$$= \frac{9 \times 2.5 \times 0.25}{2.5 \times 5} \times \frac{7}{5} = 63 \times 10^{-3} \text{ N/C}$$

8. Ans (5)



$$0.5 = \frac{kA}{L} \Delta T$$

$$\Rightarrow \Delta T = \frac{0.5 \times 1}{0.2 \times 100 \times 10^{-4}} = 250^\circ\text{C}$$

$$T_x = T_0 + \frac{250}{1} \times x$$

$$d\ell = dx \alpha (T_x - T_0)$$

$$\Delta \ell = \alpha \times 250 \int_0^1 x dx$$

$$= 2 \times 10^{-5} \times 250 \times \frac{1}{2} = 2.5 \times 10^{-3} \text{ m}$$

$$\frac{F}{A} = \frac{Y \Delta \ell}{\ell} = 2 \times 10^{11} \times \frac{2.5 \times 10^{-3}}{1}$$

$$= 5 \times 10^8 \text{ Pa}$$

9. Ans (8)

$$\rho A v^2 = mg$$

now if plate is displaced down by y

$$\rho A v = \text{constant}$$

$$v_1^2 = v^2 + 2gy$$

$$v_1 = \sqrt{v^2 + 2gy} = v \sqrt{1 + \frac{2gy}{v^2}}$$

$$F_{\text{net}} = mg - (\rho A v) v_1$$

$$= mg - mg \frac{v_1}{v} = mg \left(1 - \sqrt{1 + \frac{2gy}{v^2}} \right)$$

$$= mg \left(1 - \left(1 + \frac{2gy}{2v^2} \right) \right)$$

$$ma = -\frac{mg^2 y}{v^2}$$

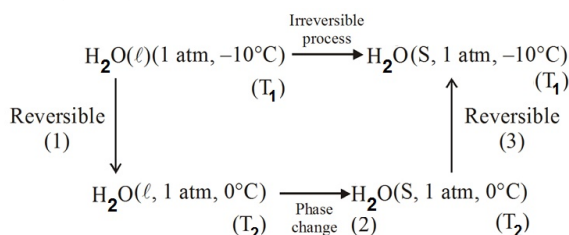
$$\frac{2\pi}{T} = \omega = \sqrt{\frac{g^2}{v^2}}$$

$$\Rightarrow T = \frac{2\pi v}{g} = \frac{2\pi \times 40}{10\pi} = 8 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,D)



For irreversible process : (T = 263 K)

$$\Delta S_{\text{surr}} = \frac{-q_{\text{irr}}}{T}$$

$$q_{\text{irr}} = \Delta H = -5.786 \text{ kJ/mole}$$

$$\therefore \Delta S_{\text{surr}} = \frac{5550}{263} = 21.1 \text{ J/K - mole}$$

Now calculate, $\Delta S_{\text{sys}}, ds_{\text{sys}} = \frac{dq_{\text{rev}}}{T}$

For reversible process,

$$\therefore \Delta S_{\text{sys}} = nC_{p(\text{water})} \ln \frac{T_2}{T_1} + \left(\frac{-\Delta H_{\text{fusion}}}{T_{f.p.}} \right) + nC_{p(\text{ice})} \ln \frac{T_1}{T_2}$$

$$= 1 \times 75 \times \ln \frac{273}{263} - \frac{600}{273} + 1 \times 30 \times \ln \frac{263}{273}$$

$$= \ln \frac{273}{263} [75 - 30] - \frac{6000}{273}$$

$$= 0.04 \times 45 - \frac{6000}{273}$$

$$= -20.18 \text{ J/K-mole}$$

$$\therefore \Delta S_{\text{sys}} = -20.18 \text{ J/K-mole}$$

$$\begin{aligned} \Delta S_{\text{total}} &= \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \\ &= (-20.18 + 21.1) \text{ J/K-mole} \\ &= 0.92 \text{ J/K-mole} \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (400.00)

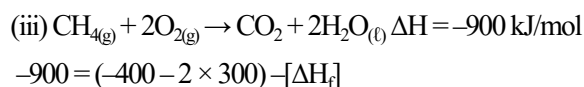
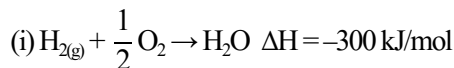


$$(-400) + 2(-300) - [-900] = -100 \Delta_f H_{\text{CH}_4(\text{g})}$$

$$-100 = 700 + 4 \times 200 - 4 \times \text{B.E.}_{\text{C-H}}$$

$$\therefore \text{B.E.}_{\text{C-H}} = 400$$

2. Ans (100.00)



$$|\Delta H_f| = 100 \text{ kJ/mol}$$

6. Ans (4.00)

(i), (iii), (iv), (vi)

PART-2 : CHEMISTRY

SECTION-II (ii)

8. Ans (6)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$\frac{y'''}{y''} = 3 \frac{y''}{y'}$$

Integrating

$$\ell \ln y'' = 3 \ell \ln y' + \ell \ln c \Rightarrow y'' = cy'^3$$

$$\frac{y''}{y'^3} = c$$

Integrating

$$-\frac{1}{2y'^2} = cx + k$$

$$= y'^2 = \frac{1}{ax + b} \quad ; a = -c, b = -k$$

$$\Rightarrow y' = \pm \frac{1}{\sqrt{ax + b}}$$

$$\Rightarrow y = \pm \frac{2\sqrt{ax + b}}{a} + \ell$$

$$\Rightarrow (y - \ell)^2 = \frac{4}{a^2} (ax + b)$$

$$\Rightarrow x = ay^2 + by + c$$

2. Ans (B)

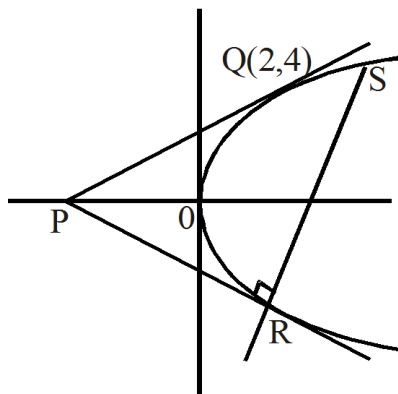
$$f(8 - a) < 0$$

$$\Rightarrow (8 - a)(8 - a - 2a + 1) < 0$$

$$\Rightarrow 3 < a < 8$$

$$p = 8 - a \Rightarrow p \in (0, 5)$$

3. Ans (B)



$$y^2 = 8x, Q(2,4) \text{ a} = 2$$

$$\Rightarrow \text{parameter of Q is } t = 1$$

$$\Rightarrow \text{parameter of R is } t = -1$$

$$\Rightarrow \text{parameter of S is } t_2 = -t_1 - \frac{2}{t_1}$$

$$\text{i.e. } t_2 = 3$$

4. Ans (B)

$$\begin{aligned} \vec{v} &= (\hat{a} \times \hat{b}) \times \hat{a} + (\hat{a} \times \hat{b}) \times \hat{b} + (\hat{a} \times \hat{b}) \times \hat{c} \\ &= \hat{b} - (\hat{a} \cdot \hat{b}) \hat{a} + (\hat{a} \cdot \hat{b}) \hat{b} - \hat{a} + (\hat{a} \cdot \hat{c}) \hat{b} - (\hat{b} \cdot \hat{c}) \hat{a} \\ &= 2(\hat{b} - \hat{a}) \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,D)

$$R^3 - 5rR^2 + 8r^2R - 4r^3 = 0$$

$$\text{Let } \frac{R}{r} = x$$

$$x^3 - 5x^2 + 8x - 4 = 0$$

$$(x-2)^2(x-1) = 0 \Rightarrow x = 2$$

$$\because x = 1 \text{ is not possible } \Rightarrow R = 2r$$

$$\Rightarrow \Delta \text{ is equilateral.}$$

6. Ans (A,C)

$$1 + 2z\bar{z} = (z^2 + 1)(\bar{z}^2 + 1) + 2(z+1)(\bar{z}+1)$$

$$(z\bar{z})^2 + 2(z+\bar{z}) + z^2 + \bar{z}^2 + 2 = 0$$

$$(z\bar{z} - 1)^2 + (z + \bar{z} + 1)^2 = 0$$

$$\Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$$

$$\text{and } z + \bar{z} + 1 = 0 \Rightarrow z + \frac{1}{z} + 1 = 0$$

$$\Rightarrow z^2 + z = -1 \Rightarrow |z^2 + z| = 1$$

7. Ans (A)

By LMVT for $f(x)$ in $x \in [1, 2]$

$\exists c_1 \in (1, 2)$ where

$$f'(c_1) = \frac{f(2) - f(1)}{2 - 1}$$

$$\because f'(x) \geq 1 \forall x \in \mathbb{R}$$

$$\Rightarrow f(2) - f(1) \geq 1 \Rightarrow f(2) \geq 2$$

By LMVT for $f(x)$ in $[2, 3]$

$\exists c_2 \in (2, 3)$ where

$$f'(c_2) = \frac{f(3) - f(2)}{3 - 2}$$

8. Ans (B,C,D)

$$f : (0, \infty) \rightarrow \mathbb{R}$$

$$F(x) = \int_0^x t f(t) dt \text{ \& } F(x^2) = x^4 + x^5$$

$$F(x^2) = \int_0^{x^2} t f(t) dt = x^4 + x^5$$

differentiating both sides.

$$x^2 f(x^2) \cdot 2x = 4x^3 + 5x^4$$

$$\Rightarrow f(x^2) = 2 + \frac{5}{2}x$$

$$\Rightarrow f(x) = 2 + \frac{5}{2}\sqrt{x} \quad x > 0$$

Clearly $f(x)$ is continuous in $x > 0$

$$f(4) = 7$$

$$\int_0^4 \left(2 + \frac{5}{2}\sqrt{x}\right) dx = \frac{64}{3}$$

but $f(x)$ has range $(2, \infty)$ hence not an onto function.

9. Ans (A,C)

$$\overrightarrow{OD} \cdot \vec{C} = 0$$

$$\{(2\lambda - 16)\hat{i} + 3\lambda\hat{j} + (2\lambda - 1)\hat{k}\} \cdot \{2\hat{i} + 3\hat{j} + 2\hat{k}\} = 0$$

$$4\lambda - 32 + 9\lambda + 4\lambda - 2 = 0 \Rightarrow \lambda = 2$$

$$\Rightarrow D \equiv (-12, 6, 3)$$

$$OD = \sqrt{189} = 3\sqrt{21}$$

$$\Rightarrow N = 21$$

D is mid-point of AB.

$$\Rightarrow B = (-4, 18, 11)$$

$$\Rightarrow a + b + c = 25$$

10. **Ans (A,C)**

$\because A - A^T$ is skew symmetric and $|A - A^T| \neq 0$

$\therefore$ order of A cannot be odd

$\Rightarrow$ order of A is 2

Now $\because AB = BA = |A|I$

$\Rightarrow B^{-1}A^{-1} = A^{-1}B^{-1}$

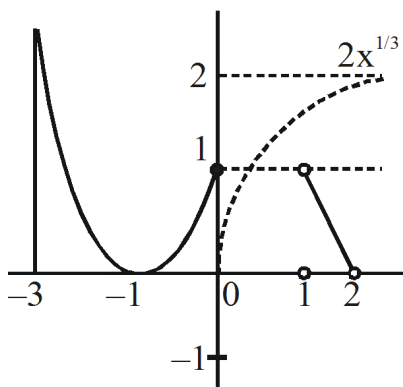
$\Rightarrow \text{adj}(B^2A^{-1}B^{-1}A) = \text{adj}(B^2B^{-1}A^{-1}A) = \text{adj}B$

$= \text{adj}(\text{adj}A) = |A|^{2-2}A = A$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (2.00)**



from graph $-1 < a < 1$

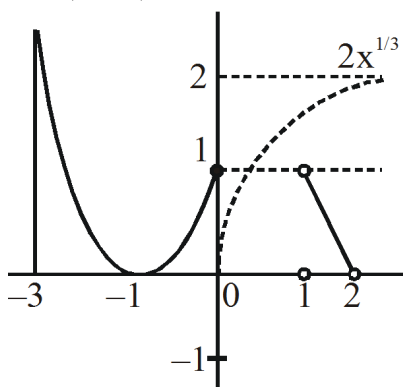
$\Rightarrow$ number of integral values of 'a' is 2

(i.e. $a = -1, 0$)

and total number of extremas of $f(x)$ is 3

(i.e. $x = -1, 0, 1$)

2. **Ans (3.00)**



from graph $-1 < a < 1$

$\Rightarrow$ number of integral values of 'a' is 2

(i.e. $a = -1, 0$)

and total number of extremas of $f(x)$ is 3

(i.e. $x = -1, 0, 1$)

3. **Ans (6.00)**

After 6 are occupied 8 vacant seats must be there and 6 seats would be in the 9 gaps.

So ${}^9C_6 6! = {}^9P_6$

4. **Ans (6.00)**

They must be seated in 3 groups of two each away from each other. These 3 groups must occupy any 3 gaps out of 9 in 9C_3 and their sitting arrangements will be

$$\frac{6!}{(2!)^3 \cdot 3!} \times {}^9C_3 \times 3! \times (2!)^3$$

5. **Ans (10.00)**

$$(x^2 + y^2 - 6x - 4y + 9) + \lambda(y - 2) = 0$$

$$\Rightarrow x^2 + y^2 - 6x - 4y + 9 = 0 \text{ (1), } y = 2 \text{ (2)}$$

$$\Rightarrow x^2 - 6x + 5 = 0 \text{ (Solving 1 \& 2)}$$

$$\therefore x = 1, 5$$

$$\Rightarrow A \equiv (1, 2) \quad B \equiv (5, 2)$$

$$x_1 + x_2 + y_1 + y_2 = 10$$

6. **Ans (2.82 or 2.83)**

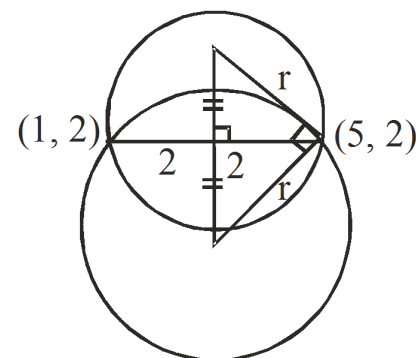
$$(x^2 + y^2 - 6x - 4y + 9) + \lambda(y - 2) = 0$$

$$\Rightarrow x^2 + y^2 - 6x - 4y + 9 = 0 \text{ (1), } y = 2 \text{ (2)}$$

$$\Rightarrow x^2 - 6x + 5 = 0 \text{ (Solving 1 \& 2)}$$

$$\therefore x = 1, 5$$

$$\Rightarrow A \equiv (1, 2) \quad B \equiv (5, 2)$$



$$\begin{aligned} \sin 45^\circ &= \frac{2}{r} = \frac{1}{\sqrt{2}} \\ &\Rightarrow r = 2\sqrt{2} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (4)**

$$xz = y^2 \text{ \& } y^2 = \frac{x+z}{2}$$

$$\Rightarrow \frac{x+z}{2} = xz$$

A.M. $\geq$ G.M.

$$\Rightarrow xz \geq \sqrt{xz} \Rightarrow (xz)^2 \geq xz$$

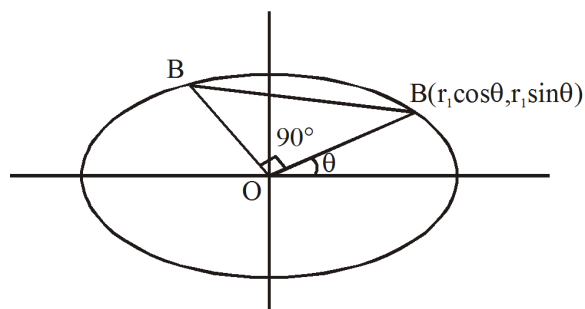
$$\because xz \neq 0 \Rightarrow xz \geq 1$$

$$\Rightarrow y^2 \geq 1 \Rightarrow y \geq 1$$

$$\text{Now, } x + 2y + z = 2y^2 + 2y$$

$$\therefore (2y^2 + 2y)_{\min} = 2 + 2 = 4$$

8. **Ans (5)**



Let $OA = r_1$ and $OB = r_2$

$$\Rightarrow A = (r_1 \cos \theta, r_1 \sin \theta)$$

$$\text{A lies on ellipse } \frac{x^2}{9} + \frac{y^2}{\lambda^2} = 1$$

$$\Rightarrow \frac{r_1^2 \cos^2 \theta}{9} + \frac{r_1^2 \sin^2 \theta}{\lambda^2} = 1$$

$$\Rightarrow \frac{1}{r_1^2} = \frac{\cos^2 \theta}{9} + \frac{\sin^2 \theta}{\lambda^2} \quad \dots\dots(1)$$

Now replace θ by $90 + \theta_1$ we get

$$\Rightarrow \frac{1}{r_2^2} = \frac{\sin^2 \theta}{9} + \frac{\cos^2 \theta}{\lambda^2} \quad \dots\dots(2)$$

$$(1) + (2) \Rightarrow \frac{14}{45} = \frac{1}{9} + \frac{1}{\lambda^2}$$

$$\Rightarrow \frac{9}{45} = \frac{1}{\lambda^2} \Rightarrow \lambda^2 = 5$$

9. **Ans (7)**

4 must be obtained on atleast one dice

x = number of dice on which 4 is obtained

required probability

$$= P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4)$$

$$= \frac{{}^4C_1 \times 3 \times 3 \times 3}{6^4} + \frac{{}^4C_2 \times 3^2}{6^4} + \frac{{}^4C_3 \times 3}{6^4} + \frac{1}{6^4}$$

$$= \frac{175}{1296}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B,C	A,B	A,B	A,C	B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	B	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.62 to 0.63	0.62 to 0.63	798.00	0.34 to 0.35	0.14	200.00
SECTION-II (ii)	Q.	7	8	9			
	A.	25	25	3			

PART-2 : CHEMISTRY

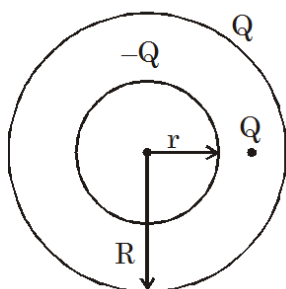
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	A,B,C,D	A,B,D	A,B,D	A,B,D	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.26	0.02	5.00	65.00	3.00	1.00
SECTION-II (ii)	Q.	7	8	9			
	A.	100	103	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C	A,B,C	B,D	A,B,C	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	6.00	1.66	10.00	25.00	11.00	0.64
SECTION-II (ii)	Q.	7	8	9			
	A.	4	12	5			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,C,D)

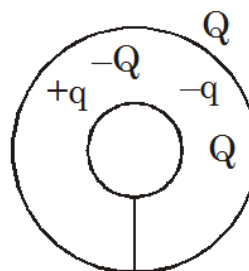


Initially

1001CJA102121085

$$V_{\text{inner}} = \frac{kQ}{L}$$

$$V_{\text{outer}} = \frac{kQ}{R}$$



finally

$$\frac{kq}{r} - \frac{k(+q)}{R} + \frac{kQ}{L} = \frac{kQ}{R}$$

HS-1/11

2. Ans (A,B,C)

	Object needle	u	Lens	u	Image needle
Convex lens	37.1 cm	-13.5 cm	50.6 cm	+27.0 cm	77.6 cm
Convex and Concave lens	37.1 cm	-13.5 cm	50.6 cm	+40.5 cm	91.1 cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{convex}}} \Rightarrow \frac{1}{27} + \frac{2}{27} = \frac{1}{f} \Rightarrow f_{\text{convex}} = 9$$

$$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f_{\text{convex}}} + \frac{1}{f_{\text{concave}}} \Rightarrow \frac{2}{27} + \frac{2}{81} = \frac{1}{f_{\text{combo}}}$$

$$\frac{1}{f_{\text{concave}}} = \frac{8}{81} - \frac{1}{9} = -\frac{1}{81} \Rightarrow f_{\text{concave}} = -27 \text{ cm}$$

$$\frac{df}{f^2} = \frac{du}{u^2} + \frac{dv}{v^2} = \frac{0.2}{729} + \frac{0.2 \times 4}{729} = \frac{1.0}{729}$$

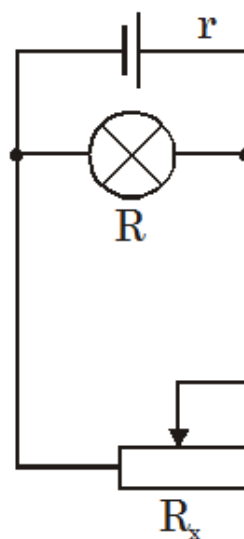
$$df \approx 0.11 \text{ cm}$$

$$\frac{df}{f^2} = \frac{du'}{u'^2} + \frac{dv'}{v'^2} + \frac{du}{u^2} + \frac{dv}{v^2}$$

$$= \frac{0.2}{81^2} \times 4 + \frac{0.2 \times 4}{27^2} + \frac{0.2 \times 4}{27^2} + \frac{0.2 \times 4}{27^2} + \frac{0.2}{27^2}$$

$$df = \frac{17}{81^2} \times 27^2 = \frac{17}{9} \text{ cm}$$

3. Ans (A,B)



$$i_B = \frac{E}{r + \frac{RR_x}{R+R_x}}$$

$$i_B = \frac{E}{r + \frac{R}{\frac{R}{R_x} + 1}}$$

$$\text{as } R_x \uparrow i_B \downarrow$$

$$i_{\text{bulb}} = i_B \times \frac{R_x}{R + R_x} = \frac{ER_x}{rR + (r + R)R_x}$$

$$= \frac{E}{\frac{rR}{R_x} + r + R}$$

$$\text{as } R_x \uparrow i_{\text{bulb}} \uparrow$$

$$i_{\text{rheostat}} = \frac{ER}{rR + (r + R)R_x}$$

$$\text{as } R_x \uparrow, i_r \downarrow$$

4. Ans (A,B)

$$\frac{dH}{dt} = \frac{mL_A}{3 \text{ sec}}$$

$$\frac{dH}{dt} = \frac{mL_B}{4 \text{ sec}}$$

$$\Rightarrow \frac{L_A}{L_B} = 0.75$$

$$\frac{dH}{dt} = m s_A \frac{dt_A}{dt} = m s_B \frac{dt_B}{dt}$$

$$\text{Solid } \frac{S_A}{S_B} = \frac{\frac{dT_B}{dt}}{\frac{dT_A}{dt}} = \frac{\frac{60}{2}}{\frac{40}{4}} = 3$$

$$\text{liquid } \frac{S_A}{S_B} = \frac{\frac{40}{2}}{\frac{20}{2}} = 2 : 1$$

5. Ans (A,C)

$$\frac{\lambda}{2} = 8\text{cm}$$

$$\lambda = 16\text{ cm}$$

$$v = f\lambda = 2000 \times \frac{16}{100} = 320\text{m/s}$$

$$\lambda' = \frac{v}{f'} = \frac{320}{1600} = 20\text{cm}$$

$$\Rightarrow \text{length} = 10\text{ cm}$$

$$\lambda = \frac{2L}{n} = 16\text{cm}$$

$$\frac{2L}{n-1} = 20\text{cm}$$

$$16n = 20n - 20$$

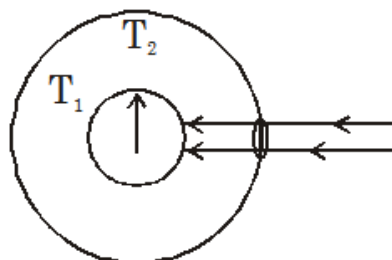
$$n = 5$$

$$\Rightarrow L = 40\text{ cm}$$

$$\frac{2L}{n-2} = \lambda' = \frac{80}{3}\text{cm}$$

$$f' = \frac{320}{80} \times 3 \times 100 = 1200\text{Hz}$$

6. Ans (B,C)



(A) A part of radiation from outer shell will also come out of hole

$$(B) \frac{dH}{dt} \text{ inner sphere} = 1600 \times A_{\text{hole}} + 4\pi \times 1^2 \sigma T_2^4 - 4\pi \times 1^2 \sigma T_1^4 = 0$$

$$(C) \frac{dH}{dt} \text{ outer shell} = 1600 A_{\text{hole}} = 4\pi \times 2^2 \sigma T_2^4$$

$$1600 \times \pi \times 10^{-4} = 4\pi \times 4 \times \frac{17}{3} \times 10^{-8} T_2^4$$

$$\frac{3}{17} \times 10^6 = T_2^4$$

$$\Rightarrow \sqrt{17} \times 10 \Rightarrow T_2 = 20\text{K}$$

$$T_1^4 = T_2^4 + \frac{1600 \times 10^{-4} \times \pi \times 3 \times 10^8}{4\pi \times 17}$$

$$T_1^4 = \frac{15}{17} \times 10^6 \Rightarrow T_1 \approx 30\text{K}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$\frac{d_I}{\frac{4}{3}} = \frac{45}{\frac{3}{2}} \Rightarrow d_I = 40$$

$$\Rightarrow u = -60\text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{60} = \frac{-3+1}{60} \Rightarrow v = -30\text{cm}$$

$$\frac{d_I}{\frac{3}{2}} = \frac{10}{\frac{4}{3}} \Rightarrow d_I = \frac{90}{8} = \frac{45}{4}$$

$$d_0 = 45 - \frac{45}{4} = \frac{135}{4}$$

$$\Rightarrow \frac{d_I}{1} = \frac{\frac{135}{4}}{\frac{3}{2}} = 22.5$$

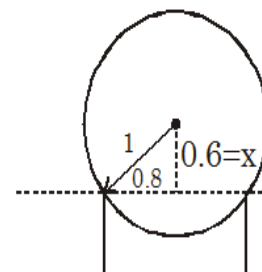
8. Ans (B)

Only magnification is at mirror.

$$\frac{h_I}{h_0} = m = \frac{-v}{u} = \frac{-(-30)}{-60} = \frac{-1}{2}$$

$$h_I = -\frac{1}{2}\text{mm}$$

9. Ans (C)



$$mg \sin \theta \times x$$

$$= I_{\text{ICR}} \alpha$$

$$= \left(\frac{2}{5} m R^2 + m \times x^2 \right) \alpha$$

$$\alpha = \frac{xg \sin \theta}{\frac{2}{5} R^2 + x^2}$$

$$a_C = x\alpha = \frac{x^2 g \sin \theta}{\frac{2}{5} R^2 + x^2}$$

$$= \frac{0.36 \times 10 \times \frac{1}{2}}{\frac{2}{5} \times 1^2 + 0.36} = \frac{1.80}{0.76}$$

10. Ans (D)

$$mg \sin \alpha - \mu mg \cos \alpha = ma_C$$

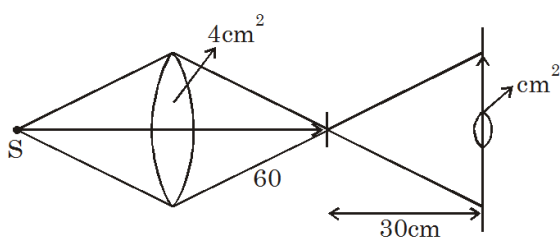
$$\mu = \tan \alpha \left(1 - \frac{x^2}{x^2 + \frac{2}{5}R^2} \right) = \frac{\frac{2}{5}R^2}{\frac{2}{5}R^2 + x^2} \tan \alpha$$

$$0.2 = \frac{0.4}{0.4 + 0.36} \tan \alpha \Rightarrow \tan \alpha = 0.38$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.62 to 0.63)



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow v = 60 \text{ cm}$$

$$i = \frac{neI\lambda}{hc} = \frac{0.09 \times \frac{100}{4\pi \times 0.3^2} \times 4 \times 10^{-4} \times \frac{\pi}{4}}{\frac{1240}{310}}$$

$$= \frac{1}{16} \times 10^{-2} \text{ A}$$

$$= 0.625 \text{ mA}$$

2. Ans (0.62 to 0.63)

$$\frac{dN}{dt} = \frac{i}{e} = \frac{0.625 \times 10^{-3}}{1.6 \times 10^{-19}}$$

$$P = k \times \frac{dN}{dt}$$

$$k_{\max} = \frac{hC}{d} - \phi = 2 \text{ eV}$$

$$p = \frac{2}{2} \times 1.6 \times 10^{-19} \times \frac{0.625 \times 10^{-3}}{1.6 \times 10^{-19}}$$

$$= 0.625 \text{ mW}$$

3. Ans (798.00)

$$n_0 = \frac{P_0 V_0}{RT}$$

$$n = \frac{PV}{RT}$$

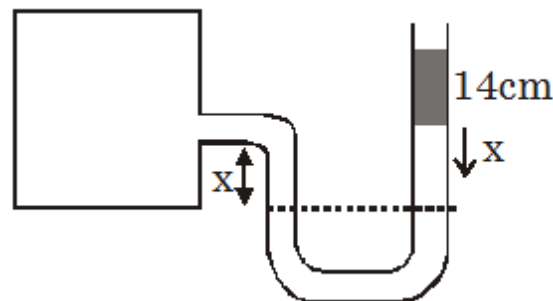
$$n_0 - n = \frac{P_0 V_0 - PV}{RT} = (75 + 20) \times 100 - \frac{(75 + 10) \times 95}{RT}$$

$$= \frac{95 \times 15 \times 14 \times 10 \times 10 \times 10^{-8}}{\frac{25}{3} \times 300}$$

$$= 57 \times 14 \times 10^{-6} \text{ moles}$$

$$= 798 \mu \text{ moles}$$

4. Ans (0.34 to 0.35)



$$P = (14\rho_{\omega}g + y\rho_{Hg}g + P_0)$$

$$PV = P_i V_i$$

$$= (P_0 + 10\rho_{Hg}g) 95A$$

$$= (P_0 + 14\rho_{\omega}g + (10 - 2x)\rho_{Hg}g) \times (95 - x)A$$

$$(75 + 10) \times 95 = (75 + 1 + 10 - 2x)(95 - x)$$

$$85 \times 95 = (86 - 2x)(95 - x)$$

$$\Rightarrow 2x^2 - 276x + 95 = 0$$

$$x = \frac{276 \pm \sqrt{276^2 - 760}}{4}$$

$$x = \frac{276 \left(1 - \left(1 - \frac{760}{276^2 \times 2} \right) \right)}{4}$$

$$= \frac{276 \times 760}{276^2 \times 8} = 0.34 \text{ cm}$$

5. Ans (0.14)

$$i_R = \frac{V}{2R} = \frac{200}{2} \text{ mA} = 0.1 \text{ A}$$

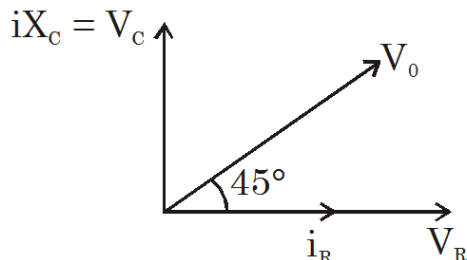
$$i_C = \frac{V}{2 \times C} = \frac{200}{2 \times \frac{1}{100 \times 10^{-5}}} = 0.1 \text{ A}$$

$$i = \sqrt{i_C^2 + i_R^2} = 0.1\sqrt{2} = 0.14 \text{ A}$$

6. Ans (200.00)

$$i_1 = i_2 = \frac{V}{Z} = \frac{200}{\sqrt{R^2 + X_C^2}}$$

$$= 0.1\sqrt{2}A$$



$$\begin{aligned}\text{Voltmeter reading} &= V_R - V_C \\ &= i_0 (R + iX_C) \\ &= 0.1\sqrt{2} \times \sqrt{2} \times 10^3 = 200 \text{ V}\end{aligned}$$

PART-1 : PHYSICS

SECTION-II (ii)

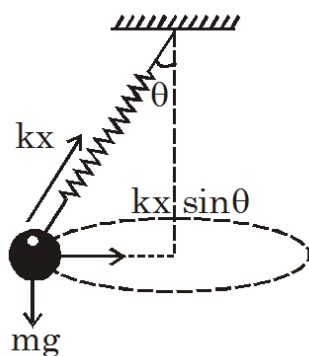
7. Ans (25)

$$m\ell = NiSB$$

$$\begin{aligned}i^2 &= \frac{0.1 \times \pi \times 10 \times \ell}{100 \times 100 \times 10^{-4} \times 4\pi \times 10^{-7} \times 1000} \\ &= 2500 \\ i &= 50A\end{aligned}$$

8. Ans (25)

$$\Delta \ell = \frac{4\pi^2 \ell_0 n^2 m}{k - 4\pi^2 n^2 m} = 25\text{cm}$$



$$\begin{aligned}kx \cos \theta &= mg \\ kx \sin \theta &= m\omega^2(\ell_0 + x) \sin \theta \\ 500 \times x \times \cos \theta &= 1 \\ 500x &= 0.1 \times 4\pi^2 \times 25 (1 + x) \\ 500x &= 100 + 100x \\ 400x &= 100 \\ x &= \frac{1}{4}m = 25\text{cm}\end{aligned}$$

9. Ans (3)

$$u_1 = \sqrt{v^2 + 2gH}$$

$$u_2 = \sqrt{2gH}$$

$$Mu_3 = mu_1 - (-mu_2) = m(\sqrt{v^2 + 2gH} + \sqrt{2gH})$$

$$\frac{mu_1^2}{2} = \frac{mu_2^2}{2} + \frac{Mu_3^2}{2}$$

$$v = \frac{\sqrt{8MmgH}}{M - m}$$

$$v = 3 \text{ m/s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,D)

$$P^\circ_{\text{solvent}} = 150 \text{ torr}$$

$$P_s = 50 \text{ torr}$$

$$\text{R.L.V.P.} = \frac{P^\circ_{\text{solvent}} - P_s}{P^\circ_{\text{solvent}}}$$

$$\text{R.L.V.P.} = \frac{150 - 50}{150}$$

$$\text{R.L.V.P.} = \frac{100}{150} = \frac{2}{3} \text{ Ans.}$$

due to H-bonding reaction between water and Ethylene glycol. It shows (-)ive deviation

$$50 \Delta H_{\text{mix}} < 0.$$

2. Ans (A,B,C,D)

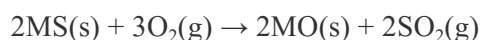
$$\text{Distance between nearest neighbour} = \frac{\sqrt{3}a}{2}$$

(= 8 in number). Distance between next nearest neighbour = a (= 6 in number)

PART-2 : CHEMISTRY

SECTION-II (i)

4. **Ans (65.00)**



The number of moles of MO and MS are equal.

We let x be the molar mass of metal. The number of moles of metal oxide is :

$$0.972\text{g} \times \frac{1 \text{ mol}}{(x + 16.00)\text{g}}$$

The number of moles of metal sulfide is :

$$1.164\text{g} \times \frac{1 \text{ mol}}{(x + 32.00)\text{g}}$$

The moles of metal oxide equal the moles of metal sulfide.

$$\frac{0.972}{(x + 16.00)} = \frac{1.164}{(x + 32.00)}$$

We solve for x.

$$0.972(x + 32.00) = 1.164(x + 16.00)$$

$$x = 65 \text{ g/mol}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

8. **Ans (103)**

Products of (i) = NO (M.wt. = 30)

(ii) = NH₃ (M.wt. = 17)

(iii) = N₂ (M.wt. = 28)

(iv) = N₂ (M.wt. = 28)

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,C)**

$$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) = (\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{b}) - \vec{a} \cdot \vec{c}$$

$$(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{c}) - \vec{a} \cdot \vec{b}$$

$$(\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{c}) - \vec{b} \cdot \vec{c}$$

$$\lambda = (\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{b}) + (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c}) + (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{c}) - (\vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c})$$

$$\text{Let } \vec{a} \cdot \vec{b} = x, \vec{b} \cdot \vec{c} = y, \vec{c} \cdot \vec{a} = z$$

$$\lambda = xy + yz + zx - (x + y + z)$$

$$\text{Also } (\vec{a} + \vec{b} + \vec{c})^2 = 3$$

$$3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 3$$

$$2(x + y + z) = 0$$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$0 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$xy + yz + zx = -(x^2 + y^2 + z^2)$$

$$\lambda = -(x^2 + y^2 + z^2) \therefore \lambda_{\max} = 0$$

$$\text{when } x = y = z = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

$$\text{So } a \perp b, b \perp c, c \perp a$$

$$\text{volume of parallelepiped} = [\vec{a} \ \vec{b} \ \vec{c}] = 1$$

$$(2\vec{a} + 3\vec{b} + 4\vec{c}) \cdot (\vec{a} \times \vec{b} + 5\vec{b} \times \vec{c} + 6\vec{c} \times \vec{a})$$

$$\Rightarrow 10[\vec{a} \ \vec{b} \ \vec{c}] + 18[\vec{b} \ \vec{c} \ \vec{a}] + 4[\vec{c} \ \vec{a} \ \vec{b}]$$

$$= 10 + 18 + 4 = 32$$

2. **Ans (A,B,C)**

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0; (\lambda - 3) = 0$$

$$\text{Now for } \lambda = 3$$

Two planes are parallel

$$\therefore \mu = 10$$

3. Ans (A,B,C)

We know that

$$|1 - \alpha_r| = \left| 1 - \cos \frac{2\pi r}{n} - i \sin \frac{2\pi r}{n} \right|$$

$$\Rightarrow |1 - \alpha_r| = \left| 2 \sin^2 \frac{\pi r}{n} - i 2 \sin \frac{\pi r}{n} \cos \frac{\pi r}{n} \right|$$

$$|1 - \alpha_r| = 2 \left| \sin \frac{\pi r}{n} \right|$$

$$\text{Similarly } |1 + \alpha_r| = 2 \left| \cos \frac{\pi r}{n} \right|$$

Now :

$$\frac{z^n - 1}{z - 1} = (z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_{n-1}) \dots (1)$$

put $z = 1$ and taking modulus both sides

$$n = |1 - \alpha_1| |1 - \alpha_2| \dots |1 - \alpha_{n-1}|$$

$$n = 2^{n-1} \left| \sin \frac{\pi}{n} \sin \frac{2\pi}{n} \dots \sin \frac{(n-1)\pi}{n} \right|$$

$$90 = 2^{89} \cdot \sin 2^\circ \cdot \sin 4^\circ \sin 6^\circ \dots \sin 178^\circ$$

$$= 2^{89} \cdot 2^{89} (\sin 1^\circ \cos 1^\circ)(\sin 2^\circ \cos 2^\circ) \dots \sin 89^\circ \cos 89^\circ$$

$$90 = 2^{89} \cdot 2^{89} \cdot \sin^2 1^\circ \sin^2 2^\circ \dots \sin^2 89^\circ$$

$$\therefore \sin 1^\circ \sin 2^\circ \sin 3^\circ \dots \sin 89^\circ = \frac{3\sqrt{10}}{289}$$

Similarly put $z = -1$ and take modulus both sides in equation(1)

$$\frac{(-1)^n - 1}{-1 - 1} = |(-1)^{n-1} (1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1})|$$

$$\left| \frac{(-1)^n - 1}{-2} \right| = 2^{n-1} \left| \cos \frac{\pi}{n} \cos \frac{2\pi}{n} \cos \frac{3\pi}{n} \dots \cos \frac{(n-1)\pi}{n} \right|$$

put $n = 45$

$$1 = 2^{44} \cos 4^\circ \cos 8^\circ \cos 12^\circ \dots \cos 172^\circ \cos 176^\circ$$

$$1 = 2^{44} \cos^2 4^\circ \cos^2 8^\circ \dots \cos^2 88^\circ$$

$$\cos 4^\circ \cos 8^\circ \dots \cos 88^\circ = \frac{1}{2^{22}}$$

$$\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11}$$

$$= \frac{\cos \left(\frac{\frac{\pi}{11} + \frac{9\pi}{11}}{2} \right) \sin \left(\frac{\frac{5\pi}{11} - \frac{7\pi}{11}}{2} \right)}{\sin \left(\frac{1}{2} \cdot \frac{2\pi}{11} \right)}$$

$$= \frac{\cos \frac{5\pi}{11} \sin \frac{5\pi}{11}}{\sin \frac{\pi}{11}} = \frac{1}{2} \frac{\sin \frac{10\pi}{11}}{\sin \frac{\pi}{11}} = \frac{1}{2}$$

4. Ans (B,D)

$$\text{Given circle } (x - 1)^2 + (y - 2)^2 = 3^2$$

$$PA = 3 \operatorname{cosec} 30^\circ = 6$$

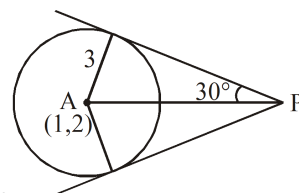
$\Rightarrow$ Locus of P is

$$(x - 1)^2 + (y - 2)^2 = 6^2$$

Radius = 6

Locus of

perpendicular tangents drawn from Q to S is director circle whose radius will be $6\sqrt{2}$



5. Ans (A,B,C)

Let n be odd integral side

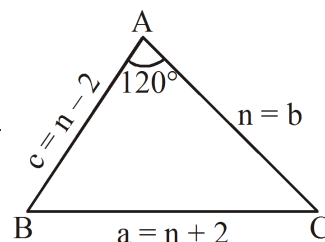
$$\cos 120^\circ = \frac{(n-2)^2 + n^2 - (n+2)^2}{2 \cdot (n-2) \cdot n}$$

$$-\frac{1}{2} = \frac{n^2 - 8n}{2n(n-2)}$$

$$\therefore -\frac{1}{2} = \frac{n-8}{2(n-2)}$$

$$\Rightarrow 2 - n = n - 8$$

$$n = 5$$



$\therefore$ sides of triangle ABC are

$$AC = 5, AB = 3, BC = 7$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{7^2 + 5^2 - 3^2}{2(7)(5)} = \frac{65}{2 \times 7 \times 5}$$

$$\cos C = \frac{13}{14}$$

$$\text{Area of } \Delta ABC = \frac{1}{2} bc \sin A = \frac{1}{2} \times 5 \times 3 \times \frac{\sqrt{3}}{2}$$

$$= \frac{15\sqrt{3}}{4}$$

$$r = \frac{\Delta}{s} = \frac{\frac{15\sqrt{3}}{4}}{\frac{1}{2}(15)} = \frac{\sqrt{3}}{2}$$

6. Ans (A,B)

$$\text{Put } x = \frac{1}{y}$$

$$\therefore dx = -\frac{1}{y^2} dy \text{ in } I_2$$

$$I_2 = \int_1^{\sin \theta} \frac{-y}{1+y^2} dy = \int_1^{\sin \theta} \frac{-x dx}{1+x^2}$$

$$I_2 = -I_1$$

$$\therefore \left| \frac{I_2}{I_1} \right| = 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$f(x) = |A - xI|$$

$$f(A) = A^2 + aA + 3I$$

$$\Rightarrow \text{tr}(A) = \text{sum of roots} = -a$$

$$\therefore a = -5$$

$$|A| = \text{product of roots} = 3$$

$$A^2 - 5A + 3I = 0$$

$$A - 5I + 3A^{-1} = 0$$

$$3A^{-1} = 5I - A$$

$$3(A^{-1})^2 = 5A^{-1} - I$$

$$3.(3A^{-1})^2 - 12A^{-1} + 3I$$

$$3(5A^{-1} - I) - 12A^{-1} + 3I$$

$$\Rightarrow 3A^{-1}$$

$$\Rightarrow 5I - A$$

$$\text{Tr}(5I - A) = \text{Tr}(5I) - \text{Tr}(A)$$

$$= 10 - 5 = 5$$

$$a_{ij} = \{1, 2, 3, 4\}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{tr}(A) = 5 \quad a + d = 5$$

$$(a, d) = (2, 3), (3, 2), (1, 4), (4, 1)$$

$$|A| = 3$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix} \text{ or } \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 3 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$$

8. Ans (B)

$$f(x) = |A - xI|$$

$$f(A) = A^2 + aA + 3I$$

$$\Rightarrow \text{tr}(A) = \text{sum of roots} = -a$$

$$\therefore a = -5$$

$$|A| = \text{product of roots} = 3$$

$$A^2 - 5A + 3I = 0$$

$$A - 5I + 3A^{-1} = 0$$

$$3A^{-1} = 5I - A$$

$$3(A^{-1})^2 = 5A^{-1} - I$$

$$3.(3A^{-1})^2 - 12A^{-1} + 3I$$

$$3(5A^{-1} - I) - 12A^{-1} + 3I$$

$$\Rightarrow 3A^{-1}$$

$$\Rightarrow 5I - A$$

$$\text{Tr}(5I - A) = \text{Tr}(5I) - \text{Tr}(A)$$

$$= 10 - 5 = 5$$

$$a_{ij} = \{1, 2, 3, 4\}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{tr}(A) = 5 \quad a + d = 5$$

$$(a, d) = (2, 3), (3, 2), (1, 4), (4, 1)$$

$$|A| = 3$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix} \text{ or } \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 3 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$$

9. Ans (A)

Probability that product is divisible by 4

$$= 1 - (\text{Divisible by 2 but not 4}) - (\text{Not divisible by 2})$$

$$= 1 - \left(\frac{1}{16} + \frac{1}{64} \right) = .9218$$

10. Ans (B)

Case-I : Product is not divisible by 2 for this we need every number to be odd.

$$\text{Probability of odd number} = \frac{1}{2}$$

$$\therefore \text{For Rolling six dice} = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$$

Case-II : Product is divisible by 2 by not by 4 we need 5 numbers to be odd and one even numbers and one number divisible by 2 but not by 4 so (2 or 6) will work.

$$\text{chance of odd number} = \frac{1}{2}$$

For divisible by 2 $\Rightarrow$ 2 or 6 will work

$$\Rightarrow \frac{2}{6} = \frac{1}{3}$$

$$\text{Required probability} = \left(\frac{1}{2}\right)^5 \cdot \frac{1}{3}$$

Even number can be rolled in any one of the six places

$$\text{Required probability} = \left(\frac{1}{2}\right)^5 \cdot \frac{1}{3} \cdot 6$$

$$\Rightarrow \frac{1}{32} \times 2 = \frac{1}{16}$$

$$\Rightarrow \frac{1}{16} = 0.0625$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (6.00)

Planes P_1 and P_2 are perpendicular

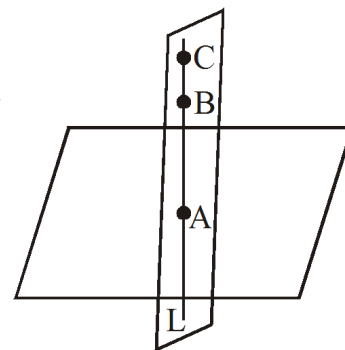
Here line L lies

in Plane P_2 and

perpendicular

to P_1 and A lies

on both planes.



$$A(-7, y_1, z_1)$$

satisfy P_1 and P_2

$$\therefore -14 - 3y_1 + z_1 = -5 \quad \dots (i)$$

$$-7 - 4y_1 - 14z_1 = 5 \quad \dots (ii)$$

Solving (i) and (ii)

$$y_1 = -3, z_1 = 0$$

Equation of line L is

$$L : \frac{x+7}{2} = \frac{y+3}{-3} = \frac{z-0}{1}$$

$$\frac{x_2+7}{2} = \frac{y_2+3}{-3} = \frac{z_2}{1} = \lambda_1$$

$$\frac{x_3+7}{2} = \frac{y_3+3}{-3} = \frac{z_3}{1} = \lambda_2$$

$$x_2 + 7 = 2\lambda_1$$

$$x_3 + 7 = 2\lambda_2$$

$$\frac{x_3 - x_2}{z_3 - z_2} = \frac{2(\lambda_2 - \lambda_1)}{(\lambda_2 - \lambda_1)} = 2$$

$$y_2 + 3 = -3\lambda_1 ; z_2 = \lambda_1$$

$$y_3 + 3 = -3\lambda_2 ; z_3 = \lambda_2$$

$$\frac{z_3 - z_2}{y_3 - y_2} = \frac{\lambda_2 - \lambda_1}{-3(\lambda_2 - \lambda_1)} = -\frac{1}{3}$$

2. Ans (1.66)

Planes P_1 and P_2 are perpendicular

Here line L lies

in Plane P_2 and

perpendicular

to P_1 and A lies

on both planes.

$A(-7, y_1, z_1)$

satisfy P_1 and P_2

$$\therefore -14 - 3y_1 + z_1$$

$$= -5 \quad \dots (i)$$

$$-7 - 4y_1 - 14z_1 = 5 \quad \dots (ii)$$

Solving (i) and (ii)

$$y_1 = -3, z_1 = 0$$

Equation of line L is

$$L : \frac{x+7}{2} = \frac{y+3}{-3} = \frac{z-0}{1}$$

$$\frac{x_2+7}{2} = \frac{y_2+3}{-3} = \frac{z_2}{1} = \lambda_1$$

$$\frac{x_3+7}{2} = \frac{y_3+3}{-3} = \frac{z_3}{1} = \lambda_2$$

$$x_2 + 7 = 2\lambda_1$$

$$x_3 + 7 = 2\lambda_2$$

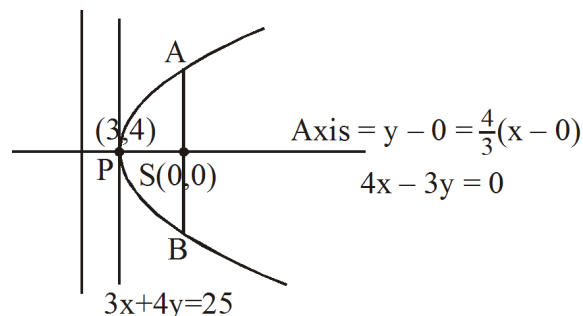
$$\frac{x_3 - x_2}{z_3 - z_2} = \frac{2(\lambda_2 - \lambda_1)}{(\lambda_2 - \lambda_1)} = 2$$

$$y_2 + 3 = -3\lambda_1 ; z_2 = \lambda_1$$

$$y_3 + 3 = -3\lambda_2 ; z_3 = \lambda_2$$

$$\frac{z_3 - z_2}{y_3 - y_2} = \frac{\lambda_2 - \lambda_1}{-3(\lambda_2 - \lambda_1)} = -\frac{1}{3}$$

3. Ans (10.00)

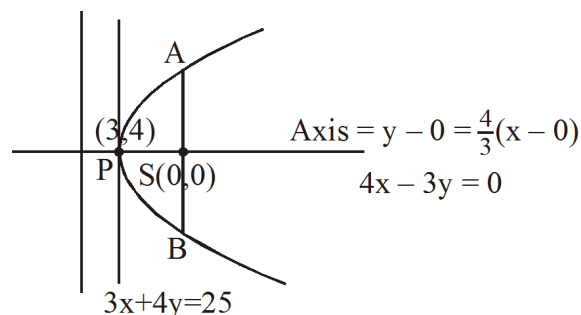


$$AB = 4PS = 4(3)$$

$$AB = 20 \text{ units}$$

$$\text{Radius of circumcircle} = 10$$

4. Ans (25.00)



$$AB = 4PS = 4(3)$$

$$AB = 20 \text{ units}$$

Let tangent at A and B meets in point Q and

line $3x + 4y = 25$ in point R and T

$$\therefore \ar \Delta QRT = \frac{1}{2} \ar \Delta APB$$

$$= \frac{1}{2} \left(\frac{1}{2} \times 20 \times 5 \right) = 25$$

5. Ans (11.00)

$$\left(1 + \frac{P(t)}{t^3} \right)^{\frac{1}{t}} = e^4$$

$$\lim_{t \rightarrow 0} \frac{1}{t} \left(\frac{P(t)}{t^3} \right) = 4$$

$$\Rightarrow P(t) = \frac{at^5 + bt^4}{t^4} = 4 \Rightarrow b = 4$$

$$P(1) = a + b = 9$$

$$\therefore a = 5$$

$$\text{so } P(x) = 5x^2 + 4x^4$$

$$y = x^2(5x^2 + 4x^4) = 5x^4 + 4x^6$$

$$\frac{dy}{dx} = 20x^3 + 24x^5$$

$$= 20(-1)^3 + 24(-5)^5 = 11$$

$$\text{Also } P(x) = 5x^2 + 4x^4$$

$$P(x) = 25x^4 + 16x^3 = x^3(25x + 16)$$

$$x = 0, x = -\frac{16}{25}$$

$$0 - \frac{16}{25} = -.64$$

$$|-.64| = .64$$

6. Ans (0.64)

$$\left(1 + \frac{P(t)}{t^3}\right)^{\frac{1}{t}} = e^4$$

$$\lim_{t \rightarrow 0} \frac{1}{t} \left(\frac{P(t)}{t^3} \right) = 4$$

$$\Rightarrow P(t) = \frac{at^5 + bt^4}{t^4} = 4 \Rightarrow b = 4$$

$$P(1) = a + b = 9$$

$$\therefore a = 5$$

$$\text{so } P(x) = 5x^2 + 4x^4$$

$$y = x^2(5x^2 + 4x^4) = 5x^7 + 4x^6$$

$$\frac{dy}{dx} = 35x^6 + 24x^5$$

$$= 35(-1)^6 + 24(-5)^5 = 11$$

$$\text{Also } P(x) = 5x^2 + 4x^4$$

$$P(x) = 25x^4 + 16x^3 = x^3(25x + 16)$$

$$x = 0, x = -\frac{16}{25}$$

$$0 - \frac{16}{15} = -.64$$

$$|-0.64| = .64$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (4)

$$f''(x) = f'(x) + e^x$$

$$f''(x) - f'(x) = e^x$$

$$\text{put } f'(x) = y$$

$$f''(x) = \frac{dy}{dx}$$

$$\frac{dy}{dx} - y = e^x$$

$$\text{I.F.} = e^{\int -1 dx} = e^{-x}$$

$$y(e^{-x}) = \int 1 dx$$

$$ye^{-x} = x + C$$

$$f'(x).e^{-x} = x + C$$

$$x = 0 \quad f'(0).1 = 0 + C$$

$$1 = C$$

$$f'(x) = xe^x + e^x$$

$$f(x) = xe^x + k$$

$$f(0) = 0 + k \therefore k = 0$$

$$f(x) = xe^x$$

$$f(2) = 2e^2$$

$$\ln \left(\frac{(f(2))^2}{4} \right) = \ln \left(\frac{4e^4}{4} \right)$$

$$= 4.00$$

8. Ans (12)

$$a + md, a + 24d, a + rd \text{ are in G.P.}$$

$$(a + 24d)^2 = (a + md)(a + rd)$$

$$\left(\frac{a}{d} + 24\right)^2 = \left(\frac{a}{d} + m\right)\left(\frac{a}{d} + r\right)$$

$$\text{Let } \frac{a}{d} = x$$

$$(x + 24)^2 = (x + m)(x + r)$$

$$x^2 + 576 + 48x = x^2 + (m + r)x + mr$$

$$48x + 576 = (m + r)x + mr$$

$$\text{Now : } 24 = \frac{2mr}{m + r}$$

$$m + r = \frac{2mr}{24}$$

$$48(x + 12) = \frac{2mr}{24} \cdot x + mr$$

$$48[x + 12] = \frac{mr}{12} [x + 12]$$

$$(x + 12) \left(48 - \frac{mr}{12} \right) = 0$$

$$x = -12 ; mr = 48 \times 12$$

$$\left| \frac{a}{d} \right| = 12$$

9. Ans (5)

$$L = \begin{cases} \lim_{x \rightarrow \pi} \frac{(x+\pi)^2(x-\pi)^2}{1 - \cos(n(\pi-x))} & ; n \text{ is odd} \\ \lim_{x \rightarrow \pi} \frac{(x^2 - \pi^2)^2}{2} & ; n \text{ is even} \end{cases}$$

$$= \begin{cases} \frac{8\pi^2}{n^2} & ; n \text{ odd} \\ 0 & ; n \text{ even} \end{cases}$$

$$\text{Let } n = 1 \Rightarrow [L] = [8\pi^2] = 78$$

$$n = 3 \Rightarrow [L] = \left[\frac{8\pi^2}{9} \right] = 8$$

$$n = 5 \Rightarrow [L] = \left[\frac{8\pi^2}{25} \right] = 3$$

$$n = 7 \Rightarrow [L] = \left[\frac{8\pi^2}{49} \right] = 1$$

$$\text{and for } n = 2, 4, 6, 8, 9, \dots [L] = 0$$